Spectrum-Conditioned Relative Reranking for MS/MS Molecule Retrieval

Abstract. Retrieving molecular structures from tandem mass spectra is difficult because mass- and formula-constrained candidate pools contain chemically similar hard negatives. We study a direct, non-generative second stage that learns to compare those candidates using the query spectrum rather than relying only on a spectrum–molecule cosine score. SpecEmbedding-Rerank first aligns a peak-sequence spectrum encoder with a molecular graph encoder and caches the complete supplied pool (at most 256 candidates). It then performs pointwise coarse scoring followed by a spectrum-conditioned relative candidate module on the coarse top 40. The relative preference is constructed as an antisymmetric pair difference and is equivariant to candidate-list permutation; no retrieval-rank feature or training-time positive insertion is used. We compare the relative model with a capacity-matched no-rank pointwise control and retain the earlier candidate-set Transformer as a secondary baseline. On MassSpecGym mass- and formula-conditioned candidate pools, both supervised rerankers improve the fixed cross-modal retriever, while the capacity-matched pointwise control is slightly stronger than the relative model at the main R@1 and MRR summaries. Thus, the paper presents the relative module as a formally defined candidate interaction mechanism and evaluates its boundary, rather than claiming a universal relation-module or external-baseline advantage. The method is closed-set and non-generative: it improves the ordering of supplied candidates but cannot recover a molecule absent from that pool.

Keywords: Data mining for bioinformatics · Tandem mass spectrometry · Molecule retrieval · Learning to rank · Multimodal representation learning

1 Introduction

MS/MS is widely used to identify small molecules in metabolomics, natural-product discovery, and related biomedical applications. An observed spectrum is only a partial, condition-dependent view of a molecular structure: different molecules can yield similar fragments, while the same molecule can produce different spectra across instruments, adducts, and collision energies. Consequently, structure annotation is naturally a large-scale data mining problem in which a query spectrum must retrieve and rank a correct structure from a chemically constrained candidate pool.

MassSpecGym provides structure-disjoint data splits and standardized mass- and formula-conditioned candidate pools for this problem [3]. Recent methods mainly address the cross-modal gap between spectra and molecules. JESTR learns a spectrum–molecule joint space with contrastive multiview coding and ranks candidates directly by cosine similarity [12,16]. GLMR instead retrieves contextual candidates, generates a refined molecule with a conditional language model, and reranks candidates by molecule–molecule similarity [19].

Both approaches leave an important data mining question open. Candidate-wise cosine scoring does not explicitly train a second-stage objective over the retrieved set, while generative retrieval adds an autoregressive decoder and makes ranking depend on the quality and validity of a generated structure. We ask whether a supervised second-stage ranker can separate hard candidates and whether a spectrum-conditioned relative comparison adds value beyond a capacity-matched pointwise scorer. The candidate pool and first-stage representations are fixed so that the comparison concerns the second-stage ranking mechanism.

Building on the peak-sequence spectrum architecture introduced by SpecEmbedding [17], we construct SpecEmbedding-Rerank, a two-stage retrieval and reranking framework. The first stage trains that spectrum tower from scratch and aligns it with a GINE molecular graph encoder. The second stage scores the full supplied pool with an absolute branch, then refines only a coarse top-K_r subset with spectrum-conditioned candidate relations. Pair preferences are formed by subtracting the score of the reversed ordered pair, which gives antisymmetry without a rank embedding. Training uses only naturally present positives; the implementation never repairs a truncated list by inserting its target. A pointwise model with the same absolute branch provides the primary capacity-controlled comparison, while the earlier candidate-set Transformer is kept as a secondary baseline.

Our contributions are:

- We formulate a direct, closed-set, non-generative second-stage learning- to-rank framework for hard candidates, using full-pool coarse scoring followed by spectrum-conditioned relative candidate comparison.
- We define a rank-free relative branch with antisymmetric pair preferences, candidate-list permutation equivariance, and no training-time positive forcing, together with a capacity-matched pointwise control.
- We characterize the framework on MassSpecGym mass- and formula-conditioned pools: supervised reranking improves the base retriever, while the current evidence does not establish an independent advantage of relative interactions over pointwise scoring.

AI-assistance disclosure. OpenAI Codex assisted throughout all sections with drafting and language revision, and during the project with code editing, experiment orchestration, and consistency audits. Reported observations and numerical results came from the stated software pipeline, not model generation. The authors reviewed all AI-assisted text and code, verified the technical claims, citations, saved artifacts, and reported results, and assume full responsibility for the manuscript's accuracy, completeness, and originality.

2 Related Work

2.1 Annotation Paradigms

Computational MS/MS annotation differs in retrieved objects and ranking evidence. Candidate-score correction predates our work: MetFusion combines in-silico fragmentation with spectral-library evidence [8]; MS2Query reranks its top 2,000 MS2DeepScore matches with a five-feature random forest [11]; and LC-MS2Struct combines an MS/MS scorer with retention order to jointly rank candidates from one LC analysis [1]. We instead rank structures in query-specific MassSpecGym pools from frozen spectrum–molecule representations, without spectral-library-neighbor or retention-time evidence [3].

Within structure-database retrieval, CSI:FingerID and MIST predict molecular fingerprints from spectra [7,9]. CMSSP and JESTR learn joint spectrum–molecule spaces; MVP extends contrastive alignment across molecular graphs, fingerprints, individual spectra, and consensus spectra [5,12,6]. These methods order candidates by predicted-fingerprint scores or cross-modal similarity. SpecEmbedding instead studies spectrum–spectrum representation learning [17]; we reuse only its peak-sequence architecture, train it from scratch against molecular graphs, and learn a complementary second-stage order within a retrieved hard-candidate list.

Other paradigms use different intermediate objects. MassFormer predicts tandem spectra from molecular graphs [18], while MSNovelist generates structures from MS/MS-derived fingerprints [15]. GLMR conditions its molecular decoder on retrieved candidates and reranks them by similarity to the generated molecule [19]. Our closed-set method does not decode a structure or return a molecule outside the first-stage list; it directly optimizes the scores of available candidates.

2.2 Learning to Rank Candidate Sets

Learning-to-rank objectives define supervision over preferences or complete lists rather than treating similarity sorting as a fixed post-processing step. RankNet learns pairwise preferences, whereas ListNet defines a probability model over a ranked list [2,4]. Ranking functions can also exploit relations among competing items. Set Transformer provides attention blocks for set-structured inputs [13], while SetRank uses self-attention to model cross-document interactions [14]. We apply these principles to cached cross-modal embeddings with explicit spectrum–candidate pair features and first-stage score and rank priors. Table 1 positions the resulting query-specific, non-generative second stage among selected related MassSpecGym approaches. The contribution is neither reranking nor self-attention itself; residual scoring and candidate interaction are evaluated design choices rather than assumed sources of gain.

Table 1. Mechanistic comparison with selected related MassSpecGym approaches.

Model	Ranking signal	Candidate interaction	Generates	Training stages
JESTR	Spectrum–candidate cosine	None at inference	No	InfoNCE; optional late-stage candidate regularization
GLMR	Generated–candidate cosine	Retrieved candidates condition the generator	Yes	Dual-path InfoNCE plus generation
Ours	Residual score from pair features and retrieval priors	Spectrum-conditioned antisymmetric pair interaction	No	Cross-modal InfoNCE plus listwise/pairwise reranking

3 Problem Formulation

Let s be an MS/MS spectrum, m^+ its correct molecule, and $\mathcal{D}_s$ a query-specific mass- or formula-conditioned candidate pool. A base retriever assigns

$$b(s, m) = \mathrm{sim}(f_s(s), f_m(m)) \tag{1}$$

to every $m \in \mathcal{D}_s$ and returns the top-K list $\mathcal{C}_s = (m_1, \ldots, m_K)$. The reranker learns scores $g(s, m_i, \mathcal{C}_s)$ such that m^+ is ordered as high as possible. Because it cannot return a molecule outside $\mathcal{C}_s$, end-to-end Recall@k is upper-bounded by

$$U_K = \frac{1}{N}\sum_{n=1}^{N} \mathbb{I}[m_n^+ \in \mathcal{C}_{s_n}]. \tag{2}$$

Define the within-list top-k accuracy conditional on first-stage coverage as

$$A_{k|K} = \frac{\sum_{n=1}^{N} \mathbb{I}[m_n^+ \in \mathcal{C}_{s_n},\, \mathrm{rank}_g(m_n^+) \le k]}{\sum_{n=1}^{N} \mathbb{I}[m_n^+ \in \mathcal{C}_{s_n}]}, \qquad \mathrm{Recall@}k = U_K A_{k|K} \le U_K. \tag{3}$$

A reranker applied to a fixed cache can therefore change only $A_{k|K}$, not U_K. We report metrics over all queries, assigning zero credit when the positive is absent rather than conditioning the primary metric on successful first-stage retrieval.

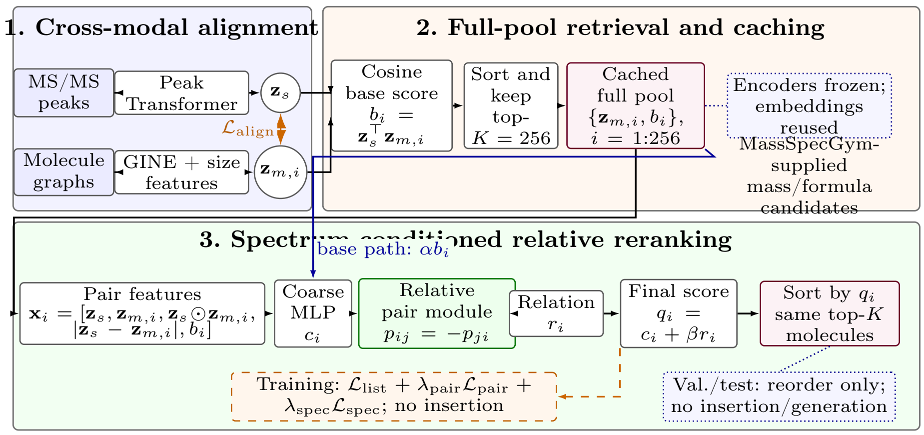

Fig. 1. Overview of SpecEmbedding-Rerank. Panel 1 aligns the spectrum Transformer and molecular GINE. Panel 2 uses their frozen, cached embeddings to produce the full candidate pool. Panel 3 applies pointwise coarse scoring followed by spectrum-conditioned relative candidate preferences on the coarse top-40; the legacy Transformer is a secondary baseline. Dashed arrows denote training-only operations. Positives are never forced into a truncated list, and no molecule is generated.

4 Method

4.1 Overview

The framework has three steps: cross-modal alignment, full-pool retrieval, and supervised coarse-to-fine relative reranking. Spectrum and molecule encoders are frozen while the reranker is trained, so their embeddings can be computed once and reused.

4.2 Cross-Modal Base Retriever

The spectrum encoder follows SpecEmbedding's peak-sequence architecture [17], which processes peak m/z values and intensities with sinusoidal m/z features and a Transformer. In our implementation, each spectrum has at most 100 tokens: a precursor-m/z token with intensity 2, followed by up to 99 fragments selected by intensity. Selected fragments retain their source-array order, and their intensities are normalized by the largest selected fragment intensity. Only these m/z–intensity tokens enter the spectrum encoder; adduct, instrument, and collision-energy metadata are not explicit inputs. We initialize this tower from scratch and train it against molecules rather than spectra. The molecular encoder is a GINE network over atom and bond features [10]. Mean-pooled graph features are augmented with explicit graph-size features. Two projection heads map the encoders to a common d-dimensional space:

$$\mathbf{z}_s = \frac{p_s(f_s(s))}{\|p_s(f_s(s))\|_2}, \qquad \mathbf{z}_m = \frac{p_m(f_m(m))}{\|p_m(f_m(m))\|_2}. \tag{4}$$

We use a bidirectional cross-modal contrastive loss that permits multiple positives. Spectra and molecules sharing a molecular label form positives, which avoids treating replicate spectra as false negatives. For a minibatch of size B, let $P(i)$ be the indices sharing the label of item i. With a learned inverse-temperature scale a, the base logits are

$$\ell_{ij} = a\,\mathbf{z}_{s_i}^\top \mathbf{z}_{m_j}. \tag{5}$$

The spectrum-to-molecule term is

$$\mathcal{L}_{s\to m} = -\frac{1}{B}\sum_{i=1}^{B}\frac{1}{|P(i)|}\sum_{j\in P(i)}\log\frac{\exp(\ell_{ij})}{\sum_{k=1}^{B}\exp(\ell_{ik})}, \tag{6}$$

and the molecule-to-spectrum term applies the same expression to $\ell^\top$. The alignment loss is their mean. The implementation parameterizes $a = \exp(u)$ and caps it at 100. At inference, the normalized dot product $b_i = \mathbf{z}_s^\top \mathbf{z}_{m_i}$ is the base score.

4.3 Candidate Construction and Training Protocol

MassSpecGym supplies target-molecule-keyed candidate lists; each query reuses the list keyed by its target molecule. The base retriever scores the complete pool of at most 256 molecules. The new coarse-to-fine reranker first scores this full pool pointwise and then applies a relation module to the coarse top $K_r = 40$ candidates. Both 256 and 40 are fixed design and compute-budget choices for this revision; the evaluation script can truncate a saved pool without inserting a positive target.

Training uses only queries whose target naturally occurs in the full 256-candidate pool. Queries without a natural positive remain in coverage statistics but do not enter the reranker loss. Candidate order is shuffled during training while each candidate's embedding and base score move together. No rank embedding is used anywhere in the new model, and the cache-compatible `base_ranks` field is ignored by its forward path. In particular, no positive is inserted into a truncated list. Missing positives at a smaller evaluation cutoff are therefore treated as coverage misses rather than repaired training examples.

4.4 Spectrum-Conditioned Relative Candidate Reranker

For candidate m_i we retain an absolute compatibility vector, but omit the retrieval rank:

$$\mathbf{x}_i = [\mathbf{z}_s,\ \mathbf{z}_{m_i},\ \mathbf{z}_s \odot \mathbf{z}_{m_i}, |\mathbf{z}_s - \mathbf{z}_{m_i}|,\ b_i]. \tag{7}$$

An absolute MLP produces a coarse score $c_i = f_{\text{abs}}(\mathbf{x}_i) + \alpha b_i$. For the coarse top-$K_r$ candidates, let $u_i = W_m \mathbf{z}_{m_i}$ and $v = W_s \mathbf{z}_s$. The relation vector ρ_{ij} contains the spectrum condition, the signed difference $u_i - u_j$, its absolute value, the base-score difference $b_i - b_j$, and the cosine similarity between u_i and u_j. A shared MLP defines an ordered preference:

$$p_{ij} = g(\rho_{ij}) - g(\rho_{ji}), \qquad p_{ij} = -p_{ji}. \tag{8}$$

The relation weights are spectrum-conditioned and normalized over valid non-self candidates. With $w_{ij} \geq 0$, the final score is

$$q_i = c_i + \beta \sum_{j \neq i} w_{ij} p_{ij}. \tag{9}$$

The relation contribution is scattered back to the full candidate pool after coarse top-K_r selection. All operations are shared across candidates and use candidate-attached masks, so jointly permuting a list and its attached features permutes the output scores in the same way. Equation 8 gives pairwise antisymmetry independently of the learned parameter values. The model uses no raw peaks, SMILES tokens, precursor mass, formula, or acquisition metadata, and no rank embedding is present in its parameter path.

The capacity-matched pointwise control uses the same absolute branch and removes only the relative aggregation. At hidden width 512, the relative model has 1,485,189 trainable parameters and the no-rank pointwise control has 1,478,690 (0.44% difference). The generic candidate-set Transformer below is retained as a secondary legacy comparison, not as the proposed mechanism.

4.5 Secondary Candidate-Set Baseline

For context, we retain the earlier rank-aware candidate-set Transformer as a secondary baseline. It uses a candidate-attached rank embedding and cross- candidate self-attention, unlike the current rank-free relative model. Its historical training-time positive forcing and configuration are not part of the formal protocol of the proposed method; further details and results are given in Appendix B.

4.6 Learning Objective

For the labeled queries $\mathcal{B}_L$ in a minibatch, let V_n be the valid candidate set and y_n the positive index. The primary objective averages listwise cross-entropy over labeled queries:

$$\mathcal{L}_{\text{list}} = -\frac{1}{|\mathcal{B}_L|} \sum_{n \in \mathcal{B}_L} \log \frac{\exp(q_{ny_n})}{\sum_{i \in V_n} \exp(q_{ni})}. \tag{10}$$

We also contrast each positive with every other valid candidate retrieved for that query. The implementation averages over all valid positive–negative pairs in the minibatch:

$$\mathcal{L}_{\text{pair}} = \frac{\sum_{n\in\mathcal{B}_L}\sum_{i\in V_n\setminus\{y_n\}}\log(1+\exp(q_{ni}-q_{ny_n}+\gamma))}{\sum_{n\in\mathcal{B}_L}(|V_n|-1)}. \tag{11}$$

The complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{list}} + \lambda_{\text{pair}}\mathcal{L}_{\text{pair}} + \lambda_{\text{spec}}\mathcal{L}_{\text{spec}}. \tag{12}$$

For a cyclicly mismatched spectrum s' from the same minibatch, the spectrum-dependency term contrasts the score of the true candidate under the correct and mismatched query:

$$\mathcal{L}_{\text{spec}} = \operatorname{softplus}\big(q(s', m^+) - q(s, m^+) + \delta\big)\,. \tag{13}$$

It is an in-batch dependency constraint, not a claim of causal identification.

The listwise term is a top-one likelihood: it separates the positive from the valid list but does not prescribe a total order among negatives. Both terms are invariant to adding the same query-specific constant to every q_i. Thus, the raw outputs are relative scores, not probabilities; probability calibration is neither performed nor evaluated.

For embedding width d, full-pool coarse scoring costs $O(Kd)$ per query and the relative branch costs $O(K_r^2 d)$ with chunked pair evaluation, rather than quadratic work over all 256 candidates. This is a structural complexity statement; measured latency, throughput, and peak memory are reported only if the fixed-hardware benchmark in the analysis artifacts has completed.

5 Experiments

5.1 Dataset and Candidate Pools

We use the published MassSpecGym split [3]: 194,119 training spectra, 19,429 validation spectra, and 17,556 test spectra. These counts total 231,104 spectra; the benchmark reports approximately 29,000 unique molecular structures. Its similarity-aware, structure-disjoint construction reduces close structural overlap between training and test molecules.

MassSpecGym supplies mass- and formula-conditioned candidate settings. The mass pool uses a precursor-mass tolerance; the formula pool assumes a known formula and is not unrestricted annotation. Frozen pools contain at most 256 candidates. The reranker is closed-set: it improves the ordering of supplied candidates but cannot recover a target absent from the pool or generate a new molecule.

Our primary evaluation follows the MassSpecGym 1.3.1 retrieval JSON order and the reference two-dimensional InChIKey identity rule. The reported table is an official-compatible candidate-order and identity evaluation using saved embeddings; it is not a fresh official-loader re-encoding or alignment retraining. A local exact-target-SMILES evaluation is retained as a sensitivity view. Further protocol details are given in Appendix A.

5.2 Implementation Details

The aligned model outputs 512-dimensional embeddings. Its spectrum Transformer has four layers, 16 heads, and width 512; the molecular GINE has four layers, hidden width 128, and dropout 0.2. No published SpecEmbedding checkpoint is loaded, so the stage that would freeze a pretrained spectrum tower is skipped; both towers are optimized end to end on the MassSpecGym training split. Alignment uses AdamW with batch size 128, learning rate and weight decay 10^{-4}, cosine scheduling, gradient-norm clipping at 1, at most 100 epochs, and patience 5. Each epoch makes ten draws per molecular label, randomly selecting an associated spectrum. Spectrum and graph augmentation are independently enabled with probability 0.5. The former removes a floor-rounded 20% of fragment peaks below normalized intensity 0.3 and jitters intensities by $\pm 15\%$; the latter applies 10% node and 10% directed edge-entry dropout. Validation disables augmentation but retains replicate-spectrum sampling under the fixed random stream. The reported reranker pilot uses one available alignment checkpoint; alignment-level variation is outside the scope of the reported reranker seed summaries.

The proposed reranker has hidden width 512, relation width 32, pair chunks of 16, and dropout 0.1. It scores the full candidate pool at $K = 256$ and applies the relative branch to its coarse top-$K_r = 40$. We use AdamW with learning rate 10^{-4}, batch size 16, at most 30 epochs, and early stopping on validation MRR with patience 5. The pairwise and spectrum-dependency weights and margins are $\lambda_{\text{pair}} = 0.2$, $\lambda_{\text{spec}} = 0.1$, $\gamma = 0.1$, and $\delta = 0.1$. The capacity-matched pointwise control uses the same absolute branch and optimization but no relation aggregation. The legacy candidate-set Transformer is retained only as a secondary comparison.

For each reported experiment, alignment checkpoints and reranker training seeds are listed with the result artifact. A reranker training seed controls initialization, data-loader order, candidate shuffling, and dropout; it is not an independent alignment replicate. Training uses only queries whose positive is naturally present in the full candidate pool. Queries without such a label are excluded from the reranker loss but retained in coverage statistics. No positive is inserted into any cache. All models are selected without test-set access. The pilot uses 5,000 training queries and three reranker seeds; this scope is a limitation of the current evidence rather than a claim of complete training or end-to-end seed stability.

5.3 Code, Data, and Artifact Availability

The anonymous supplement provides source inspection, locked environment files, automated tests, and a tiny synthetic CPU smoke test. It does not provide the MassSpecGym spectra, candidate pools, trained checkpoints, caches, or private analysis artifacts. The real-data entry point is `run_rerank_pipeline.py`; additional implementation and reproducibility notes are given in Appendix D.

Table 2. Official-compatible no-forcing full-pool pilot. Values are percentages; learned rows are means ± sample SD over reranker training seeds 42–44.

Pool	Method	R@1	R@5	R@20	R@40	MRR
Mass	SpecMolAlign (full pool)	43.78	61.10	75.55	82.15	52.19
	Capacity-matched pointwise	51.37±0.54	67.34±0.39	80.71±0.55	86.03±0.61	59.03±0.44
	Relative candidate model	50.98±0.87	67.06±0.67	80.49±0.55	86.13±0.70	58.71±0.76
Formula	SpecMolAlign (full pool)	58.49	71.50	81.81	87.30	64.79
	Capacity-matched pointwise	66.65±1.30	76.76±0.87	85.33±0.42	89.43±0.29	71.66±1.04
	Relative candidate model	66.32±0.46	76.29±0.23	85.08±0.07	89.31±0.06	71.29±0.33

5.4 Baselines and Metrics

The controlled comparisons are the aligned cosine retriever, the capacity-matched no-rank pointwise scorer, the relative reranker, and the secondary candidate-set Transformer. JESTR and GLMR provide related-work context only; their reported results are not used as a numerical comparison.

We report Recall@1, Recall@5, Recall@20, Recall@40, and mean reciprocal rank (MRR) over all 17,556 test queries. The main table uses the official-compatible candidate order and identity labels on saved embeddings. The pilot uses three reranker training seeds (42–44) on one alignment checkpoint and 5,000 training queries; means and sample SDs are seed dispersion, not independent alignment replicates. The current paper does not claim external-baseline reproduction, alignment-level stability, or end-to-end deployment performance.

5.5 Main Results

Both learned models improve on their full-pool base in both pools. Pointwise is slightly higher than relative at R@1 and MRR in the official-compatible three-seed aggregate (mass: 51.37 vs. 50.98 and 0.5903 vs. 0.5871; formula: 66.65 vs. 66.32 and 0.7166 vs. 0.7129). An evaluation-only candidate-cutoff sensitivity is included in Appendix C; it is not a separately trained model. The seed-42 paired comparison with the base retriever shows a substantial second-stage improvement for mass, while it does not establish an independent relative-module effect.

The local exact-target-SMILES sensitivity view is summarized in Appendix A and does not change the displayed conclusion.

Mechanism analysis. On the same seed-42 pilot, a base-score-only control reaches 43.78%/0.5219 (mass) and 58.49%/0.6479 (formula) for R@1/MRR, i.e., no improvement over the base. An embedding-only absolute scorer reaches 50.07%/0.5808 and 66.13%/0.7110, while adding the base-score pathway without the relative module gives 50.07%/0.5798 and 66.28%/0.7141. The full relative model gives 50.08%/0.5795 and 66.80%/0.7161. Candidate-only controls (45.00%/0.5245 and 52.73%/0.5960) are weaker than spectrum-conditioned variants. Removing molecular relations, spectrum conditioning, or antisymmetric pairing changes the two pools in different directions; these single-seed controls therefore support an information-source finding, not a causal component ranking.

JESTR and GLMR remain reported-only, not reproduced, and not directly comparable background methods. We retain their mechanism descriptions but do not reproduce their numeric table or infer a relative ranking.

Difficulty strata. In the seed-42 local query analysis, relative gains concentrate among queries that the base ranker leaves unresolved. For base ranks 2–5/6–20, relative R@1 gains over base are +42.17/+23.02 percentage points for mass and +52.72/+30.06 for formula; when the base top-1 Morgan similarity is below 0.25, the gains are +24.59/+35.92 points. Base-rank-1 queries and top-1 similarity at least 0.5 instead decline (mass -13.28/-11.86; formula -5.41/-3.65). These are descriptive seed-42 strata under the local protocol, not causal or significance evidence for the relative module.

Historical baseline boundary. Earlier rank-aware Transformer results used the former top-40 cache and training-only positive replacement. They are retained only as historical context, not as measurements of the present rank-free, no-forcing method or evidence for the relative module introduced here; further details appear in Appendix B.

6 Discussion

The new full-pool pilot separates candidate coverage from list ordering. Its full-pool upper bound is 100% by construction of the saved cache, whereas evaluation-time top-40 truncation gives a mass upper bound of 82.15%. Across three reranker seeds, relative and pointwise both improve the base; pointwise is slightly stronger at R@1/MRR, while relative is not uniformly better at any cutoff. The result supports supervised second-stage reranking as a framework component, but does not isolate an independent relative-module benefit.

The supplementary mechanism analyses show that the learned score depends on the spectrum representation, while candidate and base-score pathways also remain important. These observations are descriptive: they motivate the relative module and clarify its behavior, but do not establish a causal component gain.

The official-compatible evaluation uses the published candidate JSON and saved alignment embeddings, but does not re-encode spectra/molecules through a fresh official loader or retrain the first stage. The method is therefore evaluated within a fixed saved-embedding candidate protocol; external methods and other candidate constructions are outside the comparison.

7 Limitations and Responsible Evaluation

The reported reranker pilot uses 5,000 training queries, three reranker seeds, and one alignment checkpoint. It therefore characterizes a supervised second-stage method under a fixed representation, rather than complete alignment-level or end-to-end seed stability. The evaluation also reuses saved embeddings and supplied candidate pools; it does not claim fresh official-loader re-encoding, external-baseline reproduction, or candidate-aware alignment retraining.

The method is closed-set and formula-conditioned when the formula candidate protocol is used. It cannot recover a target absent from the candidate pool, generate a molecule outside that pool, or generalize automatically to other candidate constructions and acquisition conditions. The spectrum tower uses selected peaks and precursor m/z but no explicit adduct, instrument, or collision-energy metadata. Further protocol and reproducibility notes appear in Appendices A and D.

8 Conclusion

We introduced SpecEmbedding-Rerank, a rank-free, no-forcing second-stage reranker that combines full-pool coarse scoring with spectrum-conditioned relative candidate preferences. Across three reranker training seeds under the MassSpecGym 1.3.1- compatible candidate-order and two-dimensional-identity protocol on saved embeddings, both the relative model and its capacity-matched pointwise control improved the full-pool base in mass- and formula-conditioned retrieval. Pointwise was slightly stronger at the main R@1/MRR summaries, while supplementary spectrum-shuffle and feature controls show that both spectrum and candidate/base score pathways contribute to the learned ranking. The evidence therefore supports a carefully scoped supervised reranking framework, but does not establish an independent or generally robust benefit of the relative module. The method remains limited to the fixed saved-embedding candidate protocol.

A Evaluation Protocol and Candidate Identity

The main evaluation follows the MassSpecGym 1.3.1 retrieval candidate order and the reference two-dimensional InChIKey identity rule. The candidate sets used by the local cache agree with the supplied retrieval sets, while their ordering can differ; the primary table therefore reconstructs the reference order before scoring the saved embeddings. This is an official-compatible candidate-order and identity evaluation, not a fresh official-loader re-encoding or alignment retraining.

The tracked full-pool identity check found one identity-equivalent positive for each of the 17,556 test queries and no multiple-positive or candidate-identity collision in either candidate protocol. Exact-target-SMILES labels consequently agree with the reference identity labels on these saved pools. The local exact-target-SMILES evaluation is retained as a sensitivity view rather than a second main protocol. Formula candidates assume a known formula; both settings remain closed-set and the reranker cannot recover a target that is absent from a requested candidate cutoff.

The dataset audit also recorded six validation and three test spectrum-record matches with training records, with different target molecules. The affected validation records are excluded from the overlap-clean alignment validation configuration; the current reranker pilot instead uses its stated single alignment checkpoint and does not claim alignment-level stability. Evaluation- only candidate-cutoff sensitivities use the same saved 256-candidate cache and are not separately trained cutoff-specific models.

B Implementation and Historical Baselines

The alignment model produces 512-dimensional normalized embeddings. The spectrum tower has four Transformer layers, 16 heads and width 512; the molecular tower is a four-layer GINE with hidden width 128. The reranker uses hidden width 512, relation width 32, pair chunks of 16, dropout 0.1, $K = 256$, and $K_r = 40$. AdamW is used for both stages; the reranker uses a learning rate of 10^{-4}, batch size 16, early stopping on validation MRR, and the objective weights and margins reported in the method section.

A reranker seed controls initialization, data-loader order, candidate shuffling and dropout. The reported means and sample standard deviations over seeds 42–44 therefore describe reranker-level variation, not independent alignment replicates. Training queries without a naturally present positive remain in coverage statistics but do not enter the reranker loss. No positive is inserted into a current-method cache.

The historical Candidate-Set Transformer uses a rank embedding and generic candidate self-attention. Its earlier training-time positive forcing and rank-aware configuration are not part of the current relative/pointwise protocol. It is retained only to connect this paper with the earlier design; its larger parameter count and different training protocol prevent it from serving as the capacity-matched comparison.

C Additional Mechanism Analyses

The seed-42 information-source controls show that base-score-only scoring does not improve the base, embedding-only scoring improves it, and candidate-only controls are weaker than spectrum-conditioned variants. Removing molecular relations, spectrum conditioning or antisymmetric pairing changes the two candidate protocols in different directions. These are descriptive controls, not causal component estimates.

Difficulty analysis shows the largest relative-over-base improvements for queries whose base rank is 2–5 or 6–20, while base-rank-1 and high-similarity queries can decline. This pattern motivates the candidate-relation hypothesis but does not establish that the relation branch is universally beneficial.

The spectrum-shuffle analysis reduces the mass relative model to 45.05% Recall@1 and 0.5358 MRR, below the correctly paired result but above its base. This supports spectrum dependence while showing that candidate and base-score paths remain material. Forward-only cost measurements also show that the relative branch is more expensive than pointwise scoring; they are implementation references rather than end-to-end deployment benchmarks.

D Reproducibility and Responsible-Evaluation Notes

The current pilot uses 5,000 of 194,119 available training queries, three reranker seeds, and one alignment checkpoint. The anonymous supplement contains source inspection, locked environment files, tests and a synthetic CPU smoke test, but not the real spectra, candidate pools, caches, checkpoints, logs or private result artifacts. The smoke test is therefore not a reproduction of the scientific metrics.

JESTR and GLMR are reported-only related methods and were not reproduced under the same candidate, identity and preprocessing protocol. The paper does not claim SOTA, statistical significance, complete alignment-level stability, cross-dataset generalization or end-to-end deployment latency. These boundaries define the scope of the method result rather than additional experimental claims.

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